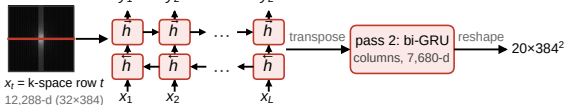


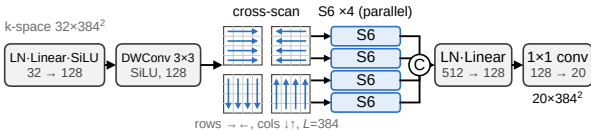
(a) bi-GRU (original ETER-Net)

668.2M (GRU stack 637.1M)

pass 1: rows — 384 steps, hidden 3,840 per direction

recurrence sequential in t ; input–hidden matrices $12,288 \times 11,520$ and $7,680 \times 11,520$ per direction**(b) SS2D (controlled substitution)**

31.2M (SSM stack 0.117M)

k-space 32×384^2 

$$S6: h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t, \quad (\Delta_t, B_t, C_t) = \text{Linear}(x_t)$$

d_inner 128, d_state 16; one weight set shared by all rows and columns